Lecture 7: Dependent Sources, Concept of Super Nodes, Super meshes

Dependent Sources:

$V_x = f(V_y)$ Voltage dependent voltage source.
$I_x = f(V_y)$ Voltage dependent Current source.
$V_x = f(I_y)$ Current dependent voltage source
$I_x = f(I_y)$ Current dependent current source.

Dependent Sources:

Dependent Sources

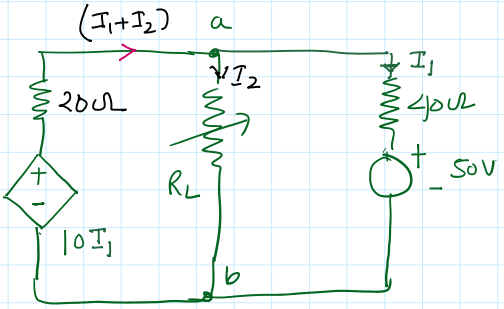
Type	Control Variable	Output	Example Expression	Application
Voltage dependent voltage source	Voltage	Voltage Source	$V_x = 5 \times V_1$	Op-amp models
Voltage dependent current source	Voltage	Current Source	$I_x = 0.2 \times V_1$	FETs, transconductance amps
Current dependent voltage source	Current	Voltage Source	$V_x = 8 \times I_1$	Transistor amplifier models
Current dependent current source	Current	Current Source	$I_x = 3 \times I_1$	#Current mirrors

Current mirror: Like current in one transistor is used as reference to produce the same current in another circuit

Example 9

Example 9: (a) First find out currents in the circuit with $R_L = 16\Omega$

(b) Find the Thevenin's equivalent circuit with R_L as load resistance. Also compute the maximum power through R_L.

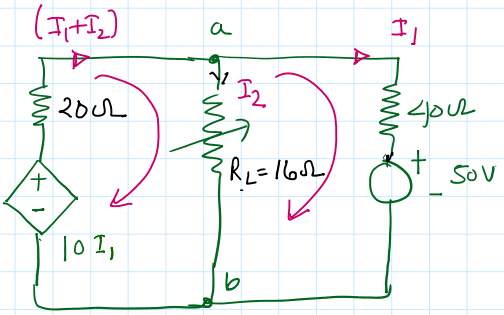


Mesh ①:

$$-10I_1 + 20(I_1 + I_2) + 16I_2 = 0$$

$$10I_1 = -36I_2$$

$$\text{or } I_2 = -\frac{5}{18}I_1 \quad ---- (1)$$



Mesh ②: $40I_1 + 50 - 16I_2 = 0$

$20I_1 - \frac{8}{16}\left(-\frac{5}{18}\right)I_1 = -25$

$20 I_1 - \frac{8}{16}\left(-\frac{5}{18}\right) I_1 = -50$ (25)

$20 I_1 + \frac{40}{18} I_1 = -25$ (20, 9)

$\left(20 + \frac{20}{9}\right) I_1 = -25$ (8)

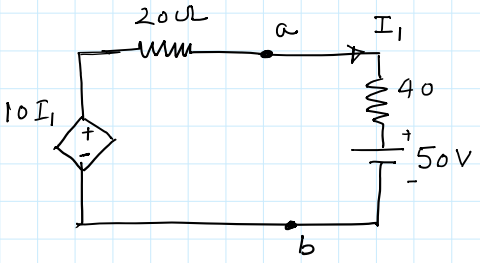
$\frac{200}{9} I_1 = -25 \Rightarrow I_1 = -\frac{9}{8}\ A = -1.125\ A$

$\Rightarrow I_2 = -\frac{5}{16} \times \left(-\frac{9}{8}\right) = \frac{5}{16}\ A$

$I_1 = -9/8\ A = -1.125\ A$

$I_2 = 5/16\ A = 0.3125\ A$

Thevenin Equivalent Circuit:



$V_{Th} = V_{ab}$,

$-10 I_1 + 20 I_1 + 40 I_1 + 50 = 0$

$\Rightarrow 50 I_1 = -50$

$I_1 = -1A$

Therefore $V_{th} = V_{ab} = 40 I_1 + 50 = -40 + 50$

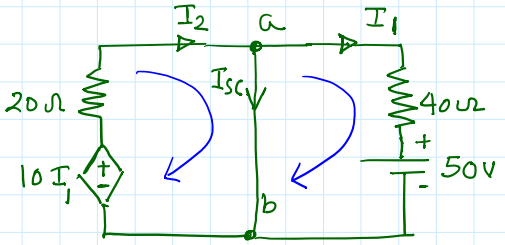
$= 10V$

$R_i = R_{ab} = R_{th}$ = Resistance into 'ab' with sources inactive.

A dependent source can not be made inactive by shorting in case of voltage and by opening in case of current.

Thus, $I_{SC} = I_N$ has to be found and then $R_{Th} = R_i = \frac{V_{Th}}{I_N} = \frac{V_{OC}}{I_{SC}}$

Let us find $I_{SC} = I_N$.



$I_{SC} = I_2 - I_1$

$= I_1/2 - I_1$

$= -I_1/2$

$-10 I_1 + 20 I_2 = 0$

$I_2 = I_1/2$

$40 I_1 + 50 = 0$

$I_1 = -\frac{5}{4}\ A$

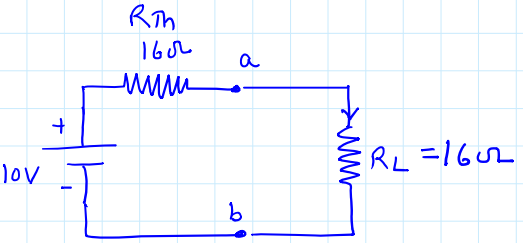
(b) The

$I_{SC} = -\frac{1}{2}\left(-\frac{5}{4}\right)$

$= \frac{5}{8}\ A$

$\therefore R_{Th} = \frac{V_{OC}}{I_{SC}} = \frac{10}{5/8} = 16\Omega$

Thus Thevenin equivalent circuit is



$\therefore I_{RL} = \frac{10}{16+16} = \frac{10}{32} = \frac{5}{16} = 0.3125\ A$

The power transfer is maximum, as $R_i = R_L$

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$$P_{L\max} = (I_{RL})^2 \times R_L = \left(\frac{5}{16}\right)^2 \times 16 = \frac{25}{16}$$

$$\boxed{P_{L\max} = 1.56\,W}$$

Q 10. In above equation, find the value of R_L so that power transferred is 0.5 W?

$$P_L = \left(\frac{10}{16+R_L}\right)^2 * R_L = \frac{100}{(16+R_L)^2} * R_L = 1/2$$

$$\Rightarrow 200 R_L = R_L^2 + 256 + 32 R_L$$

or $$R_L^2 - 168 R_L + 256 = 0$$

$$R_L = \frac{168 \pm \sqrt{(168)^2 - 4\times 256}}{2} = \frac{168 \pm \sqrt{27200}}{2}$$

$$= \frac{168 \pm 164.92}{2} = 166.46\,\Omega,\ 1.54$$

Thus $R_L = 166.46\,\Omega$, or $1.54\,\Omega$

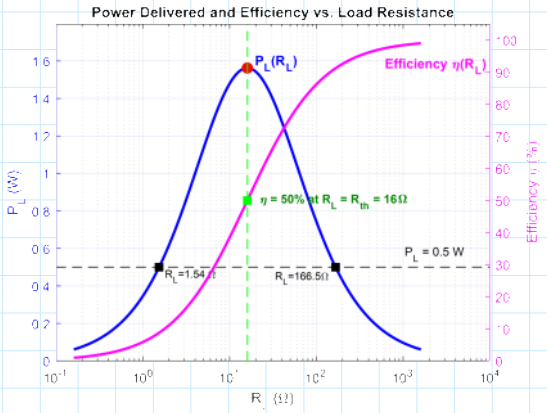
If we plot P_L versus R_L, then we see that for any value of R_L other than $R_L = R_i$, $P_L < P_{L\max}$, as depicted below.

In general $$P_L = \frac{E^2}{(R_i+R_L)^2} * R_L = \frac{E^2 * R_L}{R_i^2\left(1+\frac{R_L}{R_i}\right)^2}$$

$$P_L = \left(\frac{E^2}{R_i}\right) * \frac{(R_L|R_i)}{(1+R_L/R_i)^2}$$

$$= \left(\frac{E^2}{4R_i}\right) * \frac{4(R_L|R_i)}{(1+R_L|R_i)^2}$$

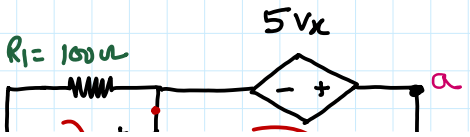
$$= P_{L\max} \cdot \frac{4*(R_L|R_i)}{(1+R_L|R_i)^2}$$



Example 10:

Consider the following circuit:

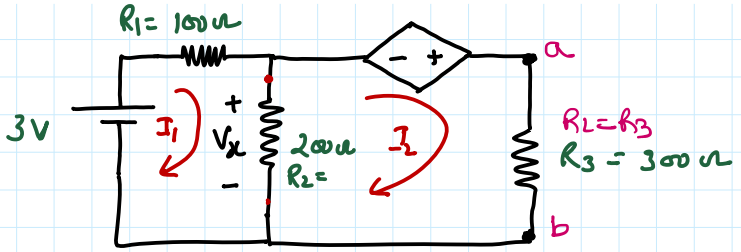
a) Find currents I_1 and I_2?



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b) Using Thevenin find current in R_3 ?

c) Find R_3 for which power delivered to it is maximum and compute maximum power.



Solution: In mesh ①

$$100 I_1 + 200 (I_1 - I_2) = 3$$

$$300 I_1 - 200 I_2 = 3 \quad \text{—— ①}$$

In mesh ②: $-5V_x + 300 I_2 + 200 (I_2 - I_1) = 0$

$$-200 I_1 + 500 I_2 = 5 V_x \quad \text{—— ②}$$

$5 V_x = 5 \times 200 (I_1 - I_2)$

$V_x = 200 (I_1 - I_2)$

$$-200 I_1 + 500 I_2 = 1000 I_1 - 1000 I_2$$

$$1200 I_1 - 1500 I_2 = 0$$

$$I_2 = \frac{1200}{1500} I_1 = \frac{4}{5} I_1$$

Substitute this value in ①

$$300 \times I_1 - 200 \times \frac{4}{5} I_1 = 3$$

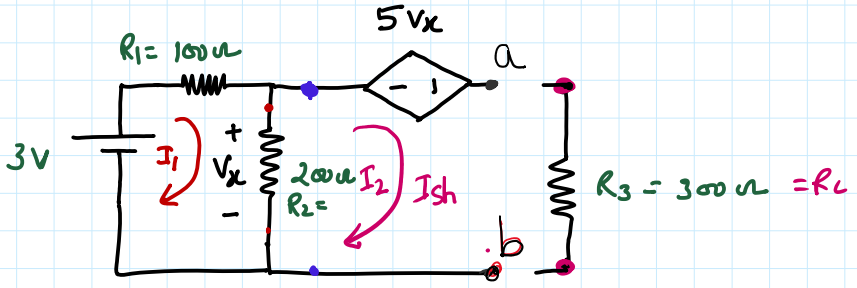
$$(300 - 160) I_1 = 3 \Rightarrow I_1 = \frac{3}{140} A = 0.0214 A = 21.4 mA$$

$$I_2 = \frac{4}{5} \times \frac{3}{140} = 0.0171 A = 17.1 mA$$

$$V_x = 200 (I_1 - I_2) = 0.8571 V$$

Find out Thevenin's Equivalent Circuit across terminal 01

$$I_1 = \frac{3}{100 + 200} = \frac{3}{300} = \frac{3}{100} = 0.01 A$$



$I_{sc} = I_2$

$V_x = 200 \times I_1 = 200 \times 0.01 = 2 V$

$V_{OC} = 5 V_x + 200 I_1 = 5 \times 2 + 200 \times 0.01 = 10 + 2 = 12 V$

Short Circuit Current:

$$100 I_1 + 200 (I_1 - I_2) = 3$$

$$300 I_1 - 200 I_2 = 3 \quad \text{—— ①}$$

$$-5 V_x + 200 (I_2 - I_1) = 0$$

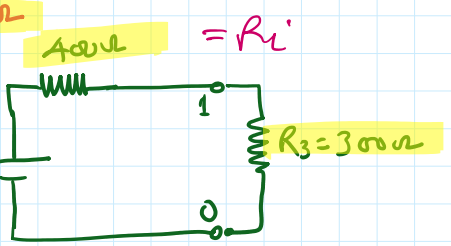
$$-5V_x + 200(I_2 - I_1) = 0$$

$$-5*200(I_1 - I_2) + 200(I_2 - I_1) = 0$$

$$I_1 = I_2$$

$$100 I_1 = 3 \Rightarrow \boxed{I_1 = \frac{3}{100} A = I_2 = I_{SC} = I_N}$$

$$R_{Th} = R_i = \frac{V_{OC}}{I_{SC}} = \frac{V_{Th}}{I_N} = \frac{12.4}{(3/100)} = 400\Omega$$



$$I_{R_3} = \frac{V_{OC}}{R_i + R_3} = \frac{V_{Th}}{R_i + R_3} = \frac{12}{400+300}$$

$$I_{R_3} = 0.0171 A$$

The value of R_3 at which maximum power is delivered

$$R_3 = R_i = 400\Omega$$

$$P_{L\,max} = \frac{V_{Th}^2}{(400+400)^2} * 400 = \frac{V_{Th}^2}{4 * R_L} = \frac{12^2}{4 \times 400} = 0.090$$

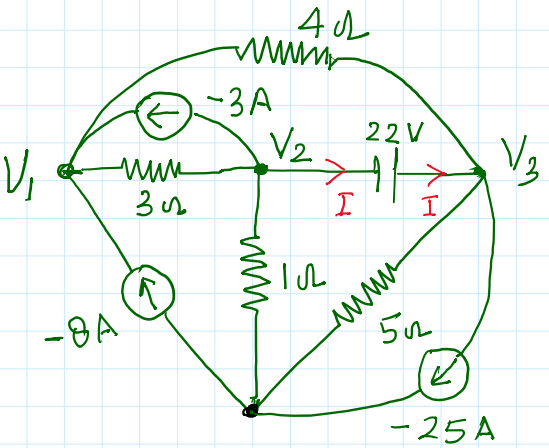
$$\boxed{P_{L\,max} = 90\ mW}$$

Example 11: Problem on circuit with supernode.

Consider the following circuit to compute V_1, V_2 & V_3.

There are 4 junctions. Using nodal methods, number of variables $= j - 1 = 4 - 3 = 3$

These are shown by V_1, V_2 & V_3.



Applying KCL at junction ①

$$\frac{V_1 - V_2}{3} + 3 + \frac{V_1 - V_3}{4} + 8 = 0$$

$$\Rightarrow \left(\frac{1}{3} + \frac{1}{4}\right) V_1 - \frac{V_2}{3} - \frac{V_3}{4} = -11$$

$$7V_1 - 4V_2 - 3V_3 = -132 \quad ---- \quad ①$$

KCL at junction ②

$$-3 + \frac{V_2 - V_1}{3} + \frac{V_2}{1} + I = 0 \quad \cdots\cdots ②$$

KCL at junction ③

$$-25 + \frac{V_3}{5} + \frac{V_3 - V_1}{4} - I = 0 \quad \cdots ③$$

Adding ② & ③ cancels I, thus nodes ② & ③ together referred as super node or super junction.

$$-3 - 25 = \frac{V_2 - V_1}{3} + \frac{V_2}{1} + \frac{V_3}{5} + \frac{V_3 - V_1}{4}$$

$$= \frac{20V_2 - 20V_1 + 60V_2 + 12V_3 + 15V_3 - 15V_1}{60}$$

$$-28 = \frac{-35V_1 + 80V_2 + 27V_3}{60}$$

$$-35V_1 + 80V_2 + 27V_3 = 1680 \quad \cdots\cdots ④$$

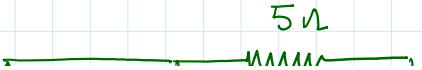
Also $V_3 - V_2 = 22V \quad \cdots\cdots ⑤$

$$\begin{bmatrix} 7 & -4 & -3 \\ -35 & 80 & 27 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -132 \\ 1680 \\ 22 \end{bmatrix}$$

$$\Rightarrow \boxed{\begin{aligned} V_1 &= 1.0714\ V \\ V_2 &= 10.5\ V \\ V_3 &= 32.5\ V \end{aligned}}$$

Example 12: Circuit solving using Super mesh concept.

KVL in mesh ①

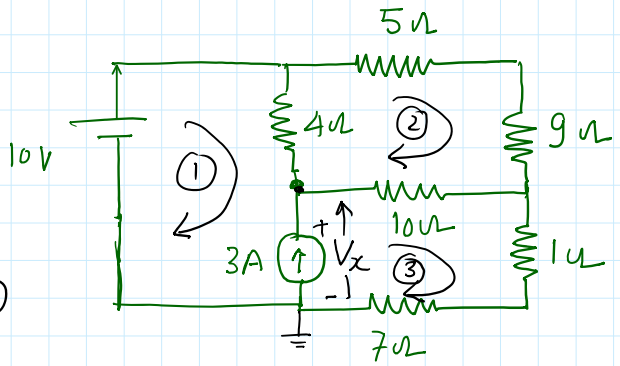


KVL in mesh ①

$$-10 + 4(I_1 - I_2) + V_x = 0 \quad \cdots ①$$

KVL in mesh ③

$$10(I_3 - I_2) + 8I_3 - V_x = 0 \quad \cdots ②$$



Combining ① & ② gives cancels V_x and thus gives KVL for supermesh ① - ③ Combined.

$$4(I_1 - I_2) + 10(I_3 - I_2) + 8I_3 = 10$$

$$4I_1 - 14I_2 + 18I_3 = 10 \quad \cdots\cdots ③$$

KCL @ mesh ②

$$13I_2 + 10(I_2 - I_3) + 4(I_2 - I_1) = 0$$

$$-4I_1 + 27I_2 - 10I_3 = 0 \quad \cdots\cdots ④$$

And $I_3 - I_1 = 3\ A$

Thus

$$\begin{bmatrix} 4 & -14 & 18 \\ -4 & 27 & -10 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 3 \end{bmatrix}$$

$$I_1 = -1.93\ A$$

$$I_2 = 0.11\ A$$

$$I_3 = 1.07\ A$$